

CSE563 — Assignment 2

Unit II: Vectors and Matrices

Maps to CO2 (demonstrate the use of vectors and matrices in image transformations). **Marks: 50.** Attempt all questions. Show every step; state matrices in full.

Part A — Short Answer and Definitions (10 marks, 2 each)

Q1. [2 marks] Define linear independence. Determine whether $\mathbf{u} = (1, 2, 3)$, $\mathbf{v} = (2, 4, 6)$, $\mathbf{w} = (1, 0, 1)$ are linearly independent.

Solution. Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent if $c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = \mathbf{0}$ forces $c_1 = \dots = c_k = 0$.

Here $\mathbf{v} = 2\mathbf{u}$, so $2\mathbf{u} - \mathbf{v} + 0\mathbf{w} = \mathbf{0}$ with non-zero coefficients. The set is **linearly dependent**. (Equivalently, the 3×3 matrix with these rows has determinant 0 and rank 2.)

Q2. [2 marks] State the condition for a matrix A to be invertible, and give two equivalent characterisations. Is $A = \begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}$ invertible?

Solution. A (square) is invertible iff $\det A \neq 0$. Equivalent statements: A has full rank; the rows (columns) are linearly independent; $A\mathbf{x} = \mathbf{0}$ has only the trivial solution; no eigenvalue is 0. $\det A = 2(3) - 6(1) = 0$, so A is **singular** (not invertible). Its second row is half the first.

Q3. [2 marks] Explain why a covariance matrix always has real eigenvalues and mutually orthogonal eigenvectors, and why this matters for PCA.

Solution. A covariance matrix $S = \tilde{X}^T \tilde{X} / (n - 1)$ satisfies $S^T = S$: it is **symmetric**. By the spectral theorem, a real symmetric matrix has all-real eigenvalues and an orthonormal set of eigenvectors.

For PCA this guarantees the principal components form a genuine orthonormal *basis*: the new axes are perpendicular, so the projected coordinates are uncorrelated and the variance decomposes additively as $\sum \lambda_i$. (It also guarantees $\lambda_i \geq 0$, since variance cannot be negative — S is positive semi-definite.)

Q4. [2 marks] Why must translation be expressed in homogeneous coordinates rather than as a plain 2×2 matrix?

Solution. A 2×2 matrix M acting on (x, y) is a *linear* map: it always sends the origin to the origin ($M\mathbf{0} = \mathbf{0}$) and cannot add a constant offset. Translation is *affine*, not linear.

Appending a third coordinate 1 and using $\begin{pmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{pmatrix}$ turns the offset into ordinary matrix

multiplication. The practical payoff: rotations, scalings and translations can all be composed into a *single* 3×3 matrix and applied in one pass.

Q5. [2 marks] State one situation where nearest-neighbour interpolation is preferable to bilinear, and explain why.

Solution. Resampling a **segmentation mask / label image** (or any categorical raster, e.g. land-cover classes, medical organ labels). The pixel values are class *identifiers*, not intensities: averaging label 2 and label 6 to get 4 invents a class that may not apply to that pixel. Nearest neighbour returns only values that exist in the source. It is also preferred when exact original pixel values must be preserved (e.g. pixel-art upscaling, or lossless integer-valued data).

Part B — Numerical Problems (26 marks)

Q6. [4 marks] For $\mathbf{u} = (2, -1, 2)$ and $\mathbf{v} = (1, 2, 2)$ compute: (a) $\mathbf{u} \cdot \mathbf{v}$; (b) $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$; (c) the angle between them; (d) the projection of $\mathbf{v}$ onto $\mathbf{u}$; (e) a unit vector along $\mathbf{u}$.

Solution. (a) $\mathbf{u} \cdot \mathbf{v} = 2(1) + (-1)(2) + 2(2) = 2 - 2 + 4 = 4$.
 (b) $\|\mathbf{u}\| = \sqrt{4 + 1 + 4} = 3$; $\|\mathbf{v}\| = \sqrt{1 + 4 + 4} = 3$.
 (c) $\cos \theta = \frac{4}{3 \times 3} = 0.4444 \Rightarrow \theta = 63.6^\circ$.
 (d) $\text{proj}_{\mathbf{u}} \mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{u}}{\|\mathbf{u}\|^2} \mathbf{u} = \frac{4}{9} (2, -1, 2) = (8/9, -4/9, 8/9)$.
 (e) $\hat{\mathbf{u}} = \frac{1}{3} (2, -1, 2) = (2/3, -1/3, 2/3)$. Check $\|\hat{\mathbf{u}}\| = \frac{1}{3} \sqrt{9} = 1$. ✓

Q7. [5 marks] Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix}$. Compute (a) AB and BA , commenting on the result; (b) $\det A$; (c) A^{-1} , verifying $AA^{-1} = I$; (d) solve $A\mathbf{x} = (5, 11)^T$.

Solution. (a) $AB = \begin{pmatrix} 0+4 & 1-2 \\ 0+8 & 3-4 \end{pmatrix} = \begin{pmatrix} 4 & -1 \\ 8 & -1 \end{pmatrix}$; $BA = \begin{pmatrix} 0+3 & 0+4 \\ 2-3 & 4-4 \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ -1 & 0 \end{pmatrix}$.
 $AB \neq BA$: matrix multiplication is **not commutative**.
 (b) $\det A = 1(4) - 2(3) = -2$.
 (c) $A^{-1} = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}$. Check: $AA^{-1} = \begin{pmatrix} -2+3 & 1-1 \\ -6+6 & 3-2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. ✓
 (d) $\mathbf{x} = A^{-1} \begin{pmatrix} 5 \\ 11 \end{pmatrix} = \begin{pmatrix} -10+11 \\ 7.5-5.5 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Check: $1(1) + 2(2) = 5$ ✓; $3(1) + 4(2) = 11$ ✓.

Q8. [5 marks] Find the eigenvalues and normalised eigenvectors of $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$. Verify your eigenvalues using the trace and determinant, and confirm that the eigenvectors are orthogonal.

Solution. Characteristic equation: $\det \begin{pmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{pmatrix} = (4-\lambda)^2 - 1 = 0 \Rightarrow (4-\lambda) = \pm 1$.

$$\lambda_1 = 5, \quad \lambda_2 = 3.$$

Check: $\text{tr } A = 8 = 5 + 3$ ✓; $\det A = 16 - 1 = 15 = 5 \times 3$ ✓.

$\lambda = 5$: $(A - 5I)\mathbf{v} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \mathbf{v} = \mathbf{0} \Rightarrow v_1 = v_2$, so $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$.

$$\lambda = 3: \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0} \Rightarrow v_1 = -v_2, \text{ so } \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1).$$

Orthogonality: $\mathbf{v}_1 \cdot \mathbf{v}_2 = \frac{1}{2}(1 - 1) = 0$. ✓ (Expected, since A is symmetric.)

Q9. [6 marks] Perform PCA by hand on the five 2-D points $(2, 1), (3, 3), (4, 3), (5, 5), (6, 8)$. (a) Centre the data. (b) Compute the 2×2 covariance matrix. (c) Find both eigenvalues and the first principal component. (d) State the percentage of variance explained by PC1. (e) Project the point $(6, 8)$ onto PC1.

Solution. (a) Centre. $\bar{x} = 20/5 = 4$; $\bar{y} = 20/5 = 4$. Centred points: $(-2, -3), (-1, -1), (0, -1), (1, 1), (2, 4)$.

(b) Covariance ($n - 1 = 4$). $\sum \tilde{x}^2 = 4 + 1 + 0 + 1 + 4 = 10$; $\sum \tilde{y}^2 = 9 + 1 + 1 + 1 + 16 = 28$; $\sum \tilde{x}\tilde{y} = 6 + 1 + 0 + 1 + 8 = 16$.

$$S = \frac{1}{4} \begin{pmatrix} 10 & 16 \\ 16 & 28 \end{pmatrix} = \begin{pmatrix} 2.5 & 4 \\ 4 & 7 \end{pmatrix}.$$

(c) Eigenvalues. $\lambda^2 - 9.5\lambda + (17.5 - 16) = 0 \Rightarrow \lambda^2 - 9.5\lambda + 1.5 = 0$.

$$\lambda = \frac{9.5 \pm \sqrt{90.25 - 6}}{2} = \frac{9.5 \pm \sqrt{84.25}}{2} = \frac{9.5 \pm 9.1788}{2}.$$

$\lambda_1 = \mathbf{9.3394}$, $\lambda_2 = \mathbf{0.1606}$. (Check: sum = 9.5 = tr S ✓.)

PC1: $(S - \lambda_1 I)\mathbf{w} = 0$: $(2.5 - 9.3394)w_1 + 4w_2 = 0 \Rightarrow w_2 = \frac{6.8394}{4}w_1 = 1.7099w_1$. Take $\mathbf{w} = (1, 1.7099)$; $\|\mathbf{w}\| = \sqrt{1 + 2.9238} = 1.9809$.

$$\mathbf{w}_1 = (0.5048, 0.8633).$$

(d) Variance explained. $\frac{9.3394}{9.5} = \mathbf{98.3\%}$ by PC1 (PC2 carries only 1.7%). The data are nearly one-dimensional.

(e) Projection of $(6, 8)$. Centred it is $(2, 4)$:

$$z = (2)(0.5048) + (4)(0.8633) = 1.0096 + 3.4532 = \mathbf{4.463}.$$

Q10. [6 marks] Two classes of 2-D samples: $C_1 = \{(2, 2), (3, 3), (4, 2), (3, 2)\}$ and $C_2 = \{(7, 6), (8, 7), (9, 6), (8, 6)\}$. (a) Compute the class means. (b) Compute S_1 , S_2 and S_W . (c) Find the LDA direction $\mathbf{w} \propto S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$. (d) Compute the threshold and classify the new point $(6, 5)$.

Solution. (a) Means. $\boldsymbol{\mu}_1 = \left(\frac{2+3+4+3}{4}, \frac{2+3+2+2}{4}\right) = (3, 2.25)$; $\boldsymbol{\mu}_2 = \left(\frac{7+8+9+8}{4}, \frac{6+7+6+6}{4}\right) = (8, 6.25)$.

(b) Scatter matrices. C_1 deviations: $(-1, -0.25), (0, 0.75), (1, -0.25), (0, -0.25)$. $\sum d_x^2 = 1 + 0 + 1 + 0 = 2$; $\sum d_y^2 = 0.0625 + 0.5625 + 0.0625 + 0.0625 = 0.75$; $\sum d_x d_y = 0.25 + 0 - 0.25 + 0 = 0$.

$$S_1 = \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix}.$$

C_2 deviations: $(-1, -0.25), (0, 0.75), (1, -0.25), (0, -0.25)$ — identical pattern, so $S_2 = \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix}$.

$$S_W = S_1 + S_2 = \begin{pmatrix} 4 & 0 \\ 0 & 1.5 \end{pmatrix}, \quad S_W^{-1} = \begin{pmatrix} 0.25 & 0 \\ 0 & 0.6667 \end{pmatrix}.$$

(c) **Direction.** $\mu_1 - \mu_2 = (-5, -4)$.

$$\mathbf{w} = S_W^{-1} \begin{pmatrix} -5 \\ -4 \end{pmatrix} = \begin{pmatrix} -1.25 \\ -2.6667 \end{pmatrix} \propto \begin{pmatrix} 1.25 \\ 2.6667 \end{pmatrix} \propto (\mathbf{3}, \mathbf{6.4}).$$

Although the means differ more in x (5) than in y (4), the classes are tighter in y (within-class scatter 1.5 vs 4), so LDA weights y more heavily.

(d) **Threshold and classification.** Using $\mathbf{w} = (1.25, 2.6667)$: $\mathbf{w}^T \mu_1 = 3.75 + 6.0 = 9.75$; $\mathbf{w}^T \mu_2 = 10 + 16.667 = 26.667$. Threshold $= (9.75 + 26.667)/2 = \mathbf{18.21}$.

New point $(6, 5)$: $\mathbf{w}^T \mathbf{x} = 7.5 + 13.33 = 20.83 > 18.21$, and since class 2 has the larger projection, the point is assigned to **class** C_2 .

Part C — Image Transformation and Interpolation (14 marks)

Q11. [6 marks] A triangle has vertices $P_1 = (1, 1)$, $P_2 = (3, 1)$, $P_3 = (1, 4)$. (a) Write the 3×3 homogeneous matrices for a scaling by $(2, 2)$ and a translation by $(3, -1)$. (b) Give the single matrix M that scales first and then translates, and apply it to all three vertices. (c) Give the matrix for the opposite order and apply it to P_1 only. Comment on the difference.

Solution. (a)

$$S = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

(b) Scale first, then translate: the right-most matrix acts first, so $M = TS$:

$$M = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Applying $(x, y) \mapsto (2x + 3, 2y - 1)$:

$$P'_1 = (5, 1), \quad P'_2 = (9, 1), \quad P'_3 = (5, 7).$$

(c) Translate first, then scale: $M' = ST = \begin{pmatrix} 2 & 0 & 6 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{pmatrix}$, i.e. $(x, y) \mapsto (2x + 6, 2y - 2)$. $P''_1 = (8, 0)$.

Comment: $P'_1 = (5, 1)$ but $P''_1 = (8, 0)$. Scaling multiplies any pre-existing translation, so the order matters — matrix multiplication is not commutative. Translating first means the offset is scaled too.

Q12. [4 marks] Rotate the point $(4, 2)$ by 90° anticlockwise about the point $(2, 2)$. Write the composite matrix and verify your result geometrically.

Solution. Sequence: translate the centre to the origin, rotate, translate back: $M = T(2, 2) R(90^\circ) T(-2, -2)$.

With $\cos 90^\circ = 0$, $\sin 90^\circ = 1$:

$$M = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 4 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Apply to $(4, 2, 1)^T$: $x' = 0(4) - 1(2) + 4 = 2$; $y' = 1(4) + 0(2) + 0 = 4$. **Result:** $(2, 4)$.

Geometric check: relative to the centre $(2, 2)$ the point is at offset $(2, 0)$ — two units due east. A 90° anticlockwise turn sends east to north, giving offset $(0, 2)$, i.e. the point $(2, 4)$. ✓ The distance from the centre is 2 before and after, as a rotation must preserve it.

Q13. [4 marks] A grayscale image has pixel values $I(0, 0) = 100$, $I(1, 0) = 150$, $I(0, 1) = 200$, $I(1, 1) = 50$ (first index = x /column, second = y /row). Estimate the intensity at $(0.4, 0.7)$ by (a) nearest-neighbour and (b) bilinear interpolation. Show the four weights and verify they sum to 1.

Solution. (a) **Nearest neighbour.** Round each coordinate: $(0.4, 0.7) \rightarrow (0, 1)$. Value = **200**.

(b) **Bilinear.** $a = 0.4$, $b = 0.7$.

Corner	Weight	Product
$I(0, 0) = 100$	$(1 - a)(1 - b) = 0.6 \times 0.3 = 0.18$	18.0
$I(1, 0) = 150$	$a(1 - b) = 0.4 \times 0.3 = 0.12$	18.0
$I(0, 1) = 200$	$(1 - a)b = 0.6 \times 0.7 = 0.42$	84.0
$I(1, 1) = 50$	$ab = 0.4 \times 0.7 = 0.28$	14.0
$\Sigma = 1.00$ ✓		134.0

Bilinear result: 134.0.

Two-step check: along x at $y = 0$: $100 + 0.4(50) = 120$; at $y = 1$: $200 + 0.4(-150) = 140$; then along y : $120 + 0.7(140 - 120) = 134$. ✓

Comment: the two methods differ by 66 grey levels here, because the nearest corner (200) happens to be an outlier among its neighbours. Bilinear smooths; nearest preserves the original value.